Death and Taxes: Inheritance Tax Planning and Unexpected Mortality

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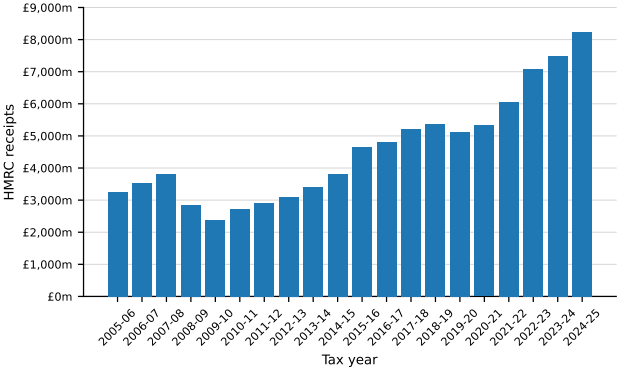
Motivation

- **Inheritance Tax (IHT) is increasingly salient in the UK**

IHT receipts, tax year 2005/2006 to tax year 2024/2025



Source: HMRC Inheritance Tax statistics

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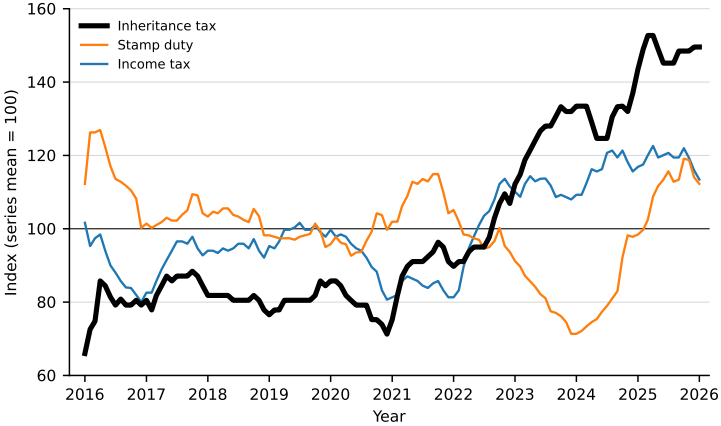

Source: Authors' calculations using Google Trends data

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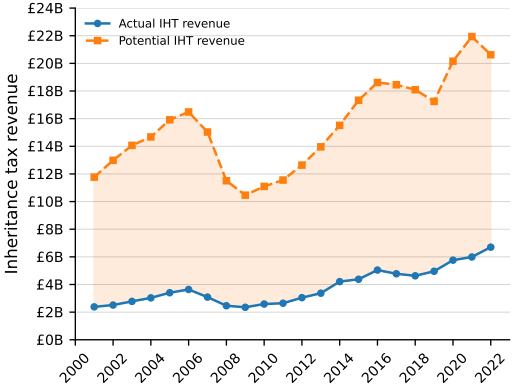
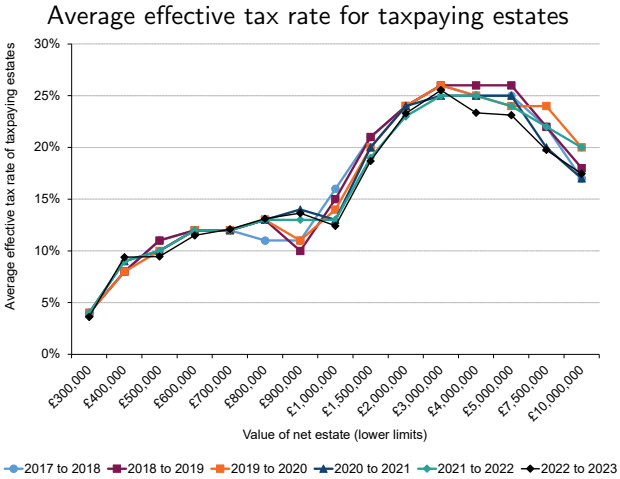


Source: Authors' calculations using HMRC and WID data

Motivation



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- ↪ **Estimates the extent of inheritance tax planning in the United Kingdom**
- ↪ **Detects the key planning mechanisms**
- ↪ **Analyzes the impact of planning on inequality (not today)**

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Methodology

- Use the first wave of COVID-19 deaths as a quasi-experimental shock in sudden deaths
- Sudden deaths reduce opportunities for tax planning
- Leverage age, gender and regional variation in excess mortality to measure the extent of tax planning in the UK in a quasi difference-in-differences setting

Literature

1. **Scant evidence on the extent of inheritance tax planning** (Wolff, 1997; Poterba, 2000; Kopczuk, 2013)
   - Quasi-experimental setting exploiting a large exogenous shock in mortality
   - Focus on the UK
   - Implications for designing and collecting a wealth tax
2. **Inheritance tax planning strategies** (Moretti and Wilson, 2023; Micó, 2024; Sturrock et al., 2025)
   - Use of IHT reliefs in the UK (Advani and Sturrock, 2023)
   - Importance of other channels for planning in the UK (*e.g.*, gifts)
3. **Estimation of wealth inequality using inheritance tax data** (Alvaredo et al., 2018, 2025; Acciari et al., 2024)

Key results

- Inheritance tax planning reduces government revenue by £3–£4.5 billion each year (55%–110% of receipts)
- Reducing the value of assets held at death (*e.g.*, through inter-vivos gifts) has a greater impact on reducing IHT liability than making strategic use of reliefs

Institutional setting

Threshold and Rate:

- IHT is levied on the total value of a decedent's estate at death + inter-vivos gifts
- *Threshold*: £325,000
- *Standard rate*: 40% on value above the threshold

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- *Spouse exemption*: transfers between spouses and civil partners are exempt
- *Residence nil-rate band*: additional allowance when passing the main residence to direct descendants
- *Business and agricultural reliefs*: reduce the taxable value of qualifying assets
- *Seven-year rule*

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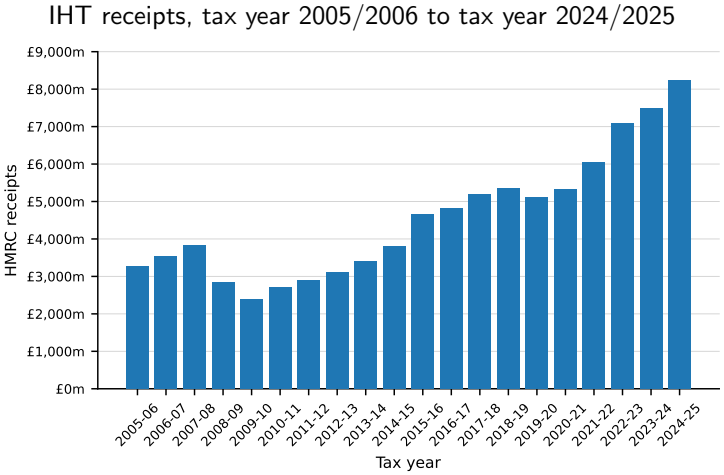
Revenues:

- £8 billion in 2024/25 ($\approx$ 1% of total tax receipts; $<$ 0.5% of GDP)

IHT in the United Kingdom



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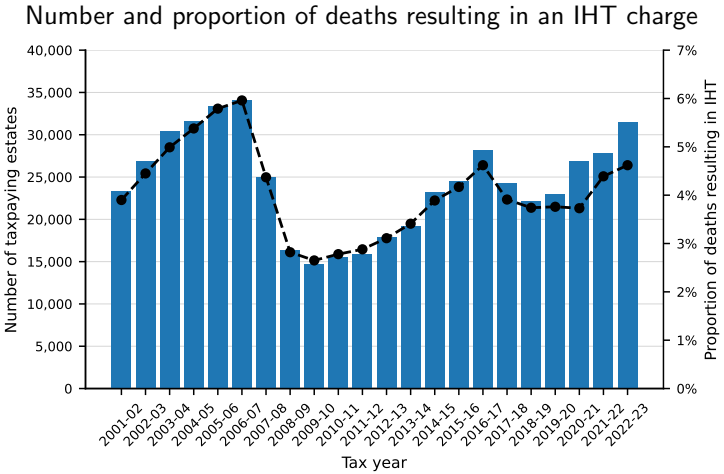

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Data

1. **Administrative tax data**
   - Inheritance tax returns (IHT400)
   - Universe of taxpaying estates and random sample of non-taxpaying estates (covering in total 40%–50% of annual deaths)
   - Covers the 2011–2022 period
   - Information on decedents:
     - Demographic characteristics (gender, age, last address, *etc.*)
     - Size and composition of the taxable estate at death
     - IHT reliefs claimed
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❷ Mortality data

- Office for National Statistics (ONS) data on excess mortality over time
- Weekly measure, available for the first COVID wave (April–May 2020)
- Disaggregated by age group, gender, and location

Descriptive statistics

Graphs

	Mean	P25	Median	P75	*N* (unw.)	*N* (wtd.)
Age	81.28	75.00	84.00	90.00	451,217	1,758,664
Female indicator	0.464	0.000	0.000	1.000	451,217	1,758,664
Married indicator	0.266	0.000	0.000	1.000	451,217	1,758,664
Other marital status indicator	0.212	0.000	0.000	0.000	451,217	1,758,664
Reliefs (£)	106,995.45	0.00	0.00	0.00	451,217	1,758,664
Tax due (£)	27,638.36	0.00	0.00	0.00	451,217	1,758,664
Total assets (£)	451,231.31	159,806.95	276,844.56	501,061.66	451,217	1,758,664
Tax rate	0.017	0.000	0.000	0.000	451,217	1,758,664
Zero tax liability indicator	0.869	1.000	1.000	1.000	451,217	1,758,664
Widowed indicator	0.522	0.000	1.000	1.000	451,217	1,758,664
Unexpected mortality risk (April)	0.461	0.408	0.473	0.514	451,217	1,758,664
Unexpected mortality risk (May)	0.264	0.205	0.300	0.335	451,217	1,758,664

Methodology I

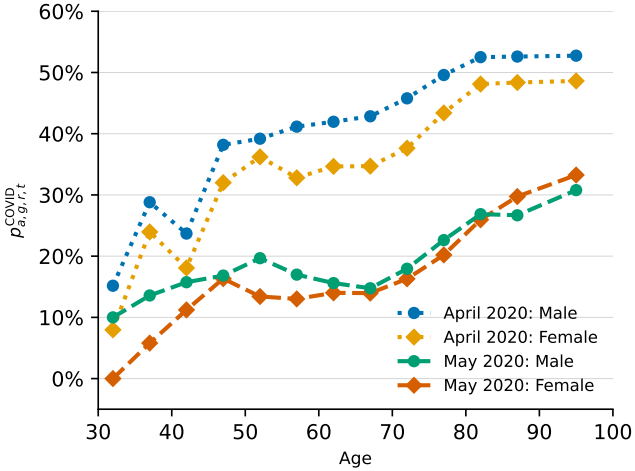
Identify unexpected deaths

- Use the first wave of COVID-19 deaths in 2020 as an exogenous mortality shock
- Increases the probability of dying unexpectedly, limiting opportunities for estate planning
- Compute the conditional probability that an individual who died in April–May 2020 died unexpectedly, for each age group a, gender g, region r, and month t:

$$p^{\text{COVID}}_{a,g,r,t} = \frac{\text{Actual_Deaths}_{a,g,r,t} - \text{Predicted_Deaths}_{a,g,r,t}}{\text{Actual_Deaths}_{a,g,r,t}}$$

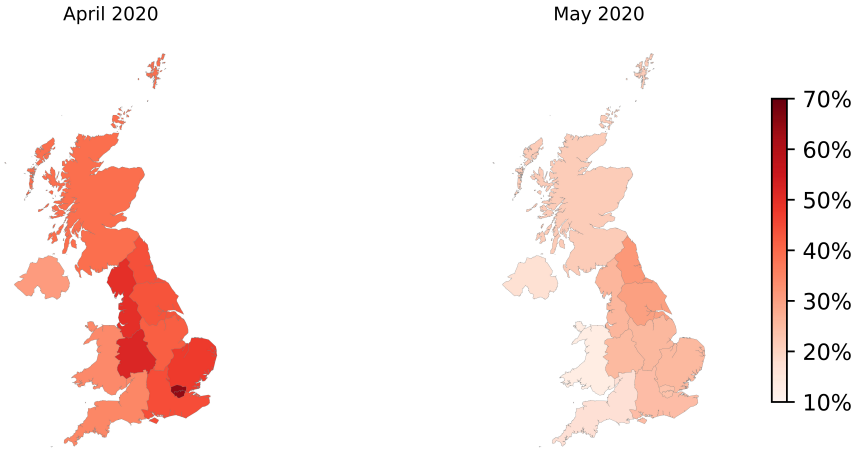
- Set $p^{\text{COVID}}_{a,g,r,t} = 0$ outside the first wave of COVID-19 and for decedents younger than 30 or older than 80 years (in our preferred specification)

Conditional unexpected death probability by age and gender



Source: Authors' calculations using data from the ONS

Conditional unexpected death probability by region



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Methodology II

Quasi difference-in-differences setting

- Construct a panel by averaging key variables from individual tax returns within each age group (a) and gender (g) for each month (t)
- Compute effect of unexpected deaths on IHT in 2020:

$$y_{a,g,t} = \alpha + \delta \cdot p^{\text{COVID}}_{a,g,t} + \theta_t + \kappa_{a,g} + \epsilon_{a,g,t}$$

  - $y_{a,g,t}$: total wealth, tax liability, effective IHT rate, or reliefs claimed
  - θ_t: period (month) fixed effects
  - $\kappa_{a,g}$: age-by-gender fixed effects

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- Construct a panel by averaging key variables from individual tax returns within each age group (a) and gender (g) for each month (t)
- Also estimating placebo effects for previous years:

$$y_{a,g,t} = \alpha + \sum_{y=2016}^{2020} \delta_y \cdot p_{a,g,t}^{\text{COVID}} + \theta_t + \kappa_{a,g} + \epsilon_{a,g,t}$$

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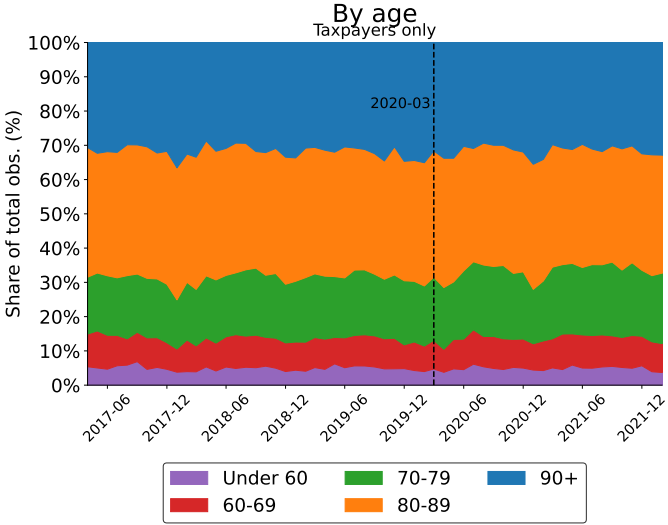
- Expect the estimated placebo effects $\hat{\delta}_{2016}$–$\hat{\delta}_{2019}$ to be null

Composition of taxpaying estates remains stable during COVID-19



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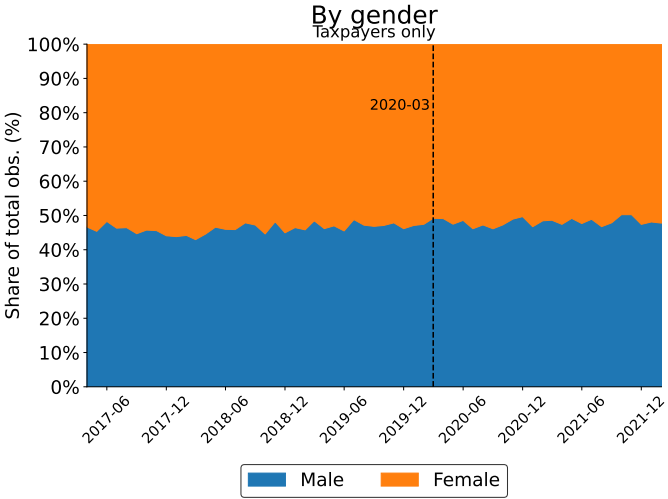

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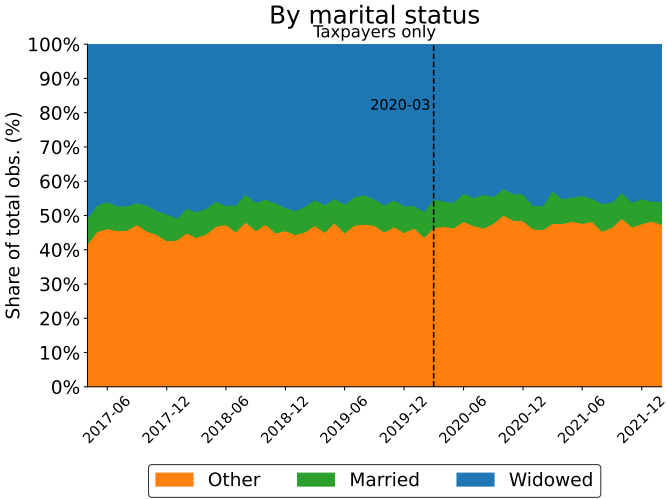

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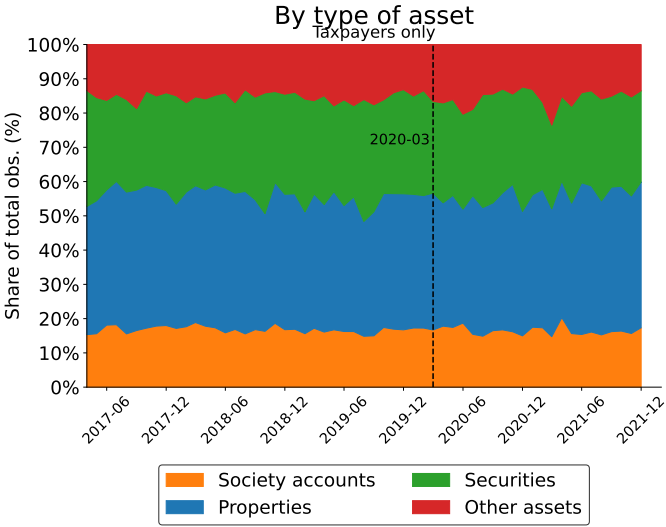

Wealth gradient in excess mortality

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  - During COVID: steeper negative income gradient (Brandily et al., 2021; Calderón-Larrañaga et al., 2020; Chen et al., 2020)
  - Mediated by strong geographic location effects (Decoster et al., 2021)

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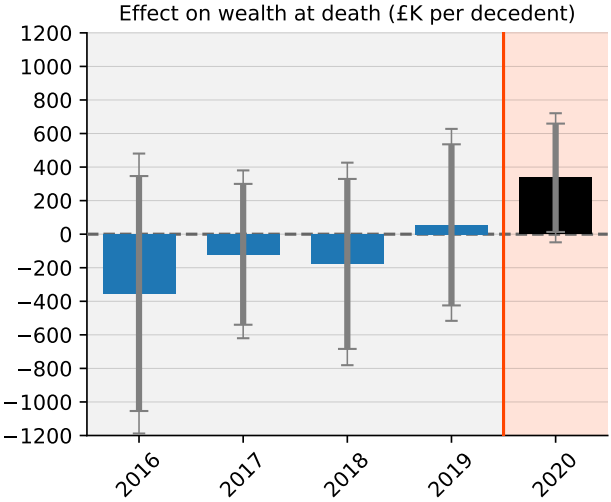
↪ **We likely estimate a lower-bound of inheritance planning**

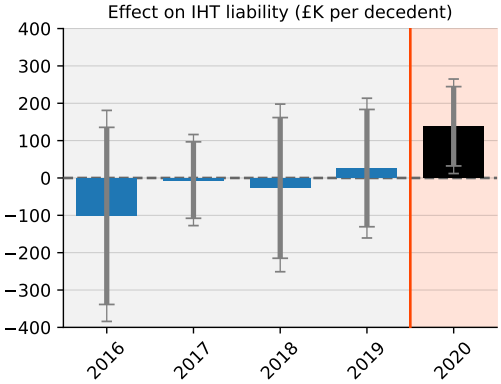
Results summary

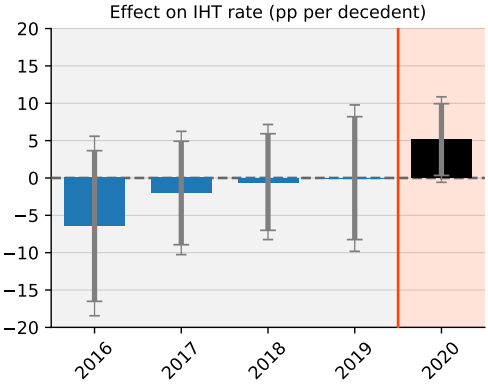
	Total Assets				Tax Rate				Tax Due			
	Ages 1–90		Ages 30–80		Ages 1–90		Ages 30–80		Ages 1–90		Ages 30–80	
δ	601310.6**	579041.1**	352119.8*	332723.1*	0.0694**	0.0678*	0.0532*	0.0519*	193622.6**	191910.1**	142123.5**	140628.7**
	(240847.0)	(238836.6)	(196164.7)	(195984.3)	(0.0340)	(0.0341)	(0.0289)	(0.0289)	(77917.5)	(77855.7)	(65205.0)	(65197.7)
β_{London}		188674.8***		188944.1***		0.0135*		0.0135*		14508.6		14560.9
		(65184.0)		(65106.3)		(0.00793)		(0.00792)		(21183.9)		(21158.5)
Adj. R^2	0.228	0.233	0.228	0.233	0.325	0.326	0.324	0.326	0.197	0.197	0.197	0.197
N	3490	3490	3490	3490	3490	3490	3490	3490	3490	3490	3490	3490

Results: The effect of unexpected death probability on wealth at death

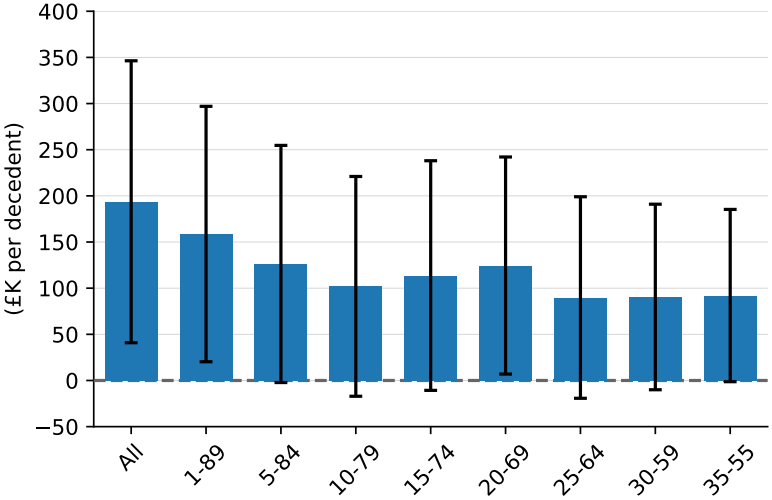


Results: The effect of unexpected death probability on IHT liability and rate





Results: Robustness to treated age group definition



Mechanisms: Estimated effect on wealth components at death

	Total Assets	Savings	Properties	Farm	Business	Other Assets
δ	352119.8*	365042.3***	-11967.6	-11841.2	-29229.7*	38663.9
	(196164.7)	(133210.3)	(76004.4)	(7347.9)	(16798.8)	(40666.5)
Adj. R^2	0.23	0.21	0.25	0.09	0.01	0.03
N	3490	3490	3490	3490	3490	3490

Mechanisms: Estimated effect on claimed reliefs

	Total Reliefs	Business Reliefs	Property Reliefs	Other Reliefs
δ	-37812.9	-30296.0**	-7,624.5	107.6
	(29175.4)	(14850.8)	(6,789.5)	(21338.3)
Adj. R^2	0.03	0.01	0.25	0.02
N	3490	3490	3490	3490

How much revenue is lost to IHT planning?

A simple model:

$$T_o^i = T_t^i - \beta \cdot \pi \cdot \left(1 - p^i\right)$$

- T_o^i: observed tax due
- T_t^i: true tax due
- β: average amount of tax reduced by planning
- π: probability that an individual i planned their estate
- p^i: probability that i died unexpectedly

How much revenue is lost to IHT planning?

A simple model:

Rearranging:

$$T_o^i = \left(T_t^i - \beta \cdot \pi\right) + \beta \cdot \pi \cdot p^i \,,$$

which closely resembles our regression model:

$$y_i = \alpha_i + \delta p_i$$

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The total amount of tax lost can be found by aggregating over all individuals:

$$\Delta T = \sum_i (T_t^i - T_o^i) = \sum_i \beta \cdot \pi \cdot \left(1 - p^i\right) = \sum_i \beta \cdot \pi = \sum_i \hat{\delta} = \hat{\delta} N \,,$$

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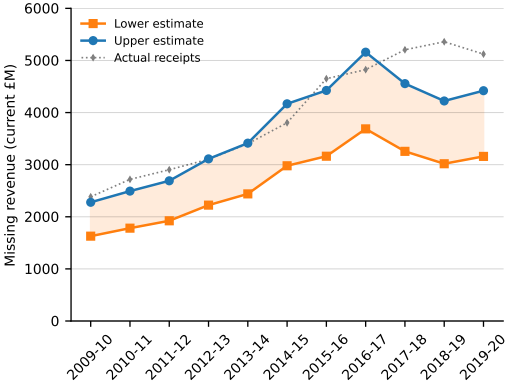
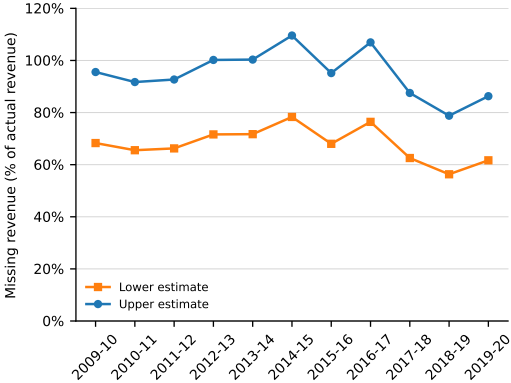
For 2019:

- Taxpaying estates: 22,500
- Estimated $\hat{\delta}$: £140,000

$\rightarrow$ Total revenues lost: $\approx$ £3 billion, or 65% of total IHT revenues over the year

Extended to other years





Source: authors' calculations from HMRC data

Conclusion

This paper

- Uses unexpected deaths as a quasi-experimental shock to planning
- To identify the extent of IHT planning in the UK

Findings

- IHT planning is substantial: reduces revenues by £3–£4.5 billion each year (55%–110% of receipts)
- IHT planning primarily operates through reductions in taxable estate at death (possibly through inter-vivos gifts)
- rather than via the use of tax-deductible reliefs

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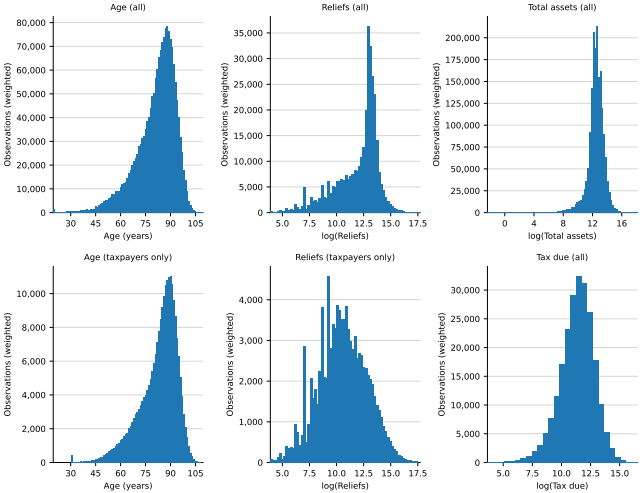
Descriptive statistics

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